SYEDA BATOOL HASSAN

AI Engineer | Data Scientist | Data Analyst | MLOps | Python
syedabatoolhassan123@gmail.com | 03330204117 | Karachi, Pakistan
LinkedIn: Syeda batool Hassan | LinkedIn | GitHub: SyedaBatool123 · GitHub

PROFESSIONAL SUMMARY

Highly motivated and results-driven Data Scientist with hands-on experience in designing, developing, and deploying end-to-end Artificial Intelligence and Machine Learning solutions. Demonstrated expertise in Machine Learning, Data Science, Predictive Analytics, Intelligent Automation, MLOps fundamentals, REST API Development & Integration, CI/CD, Explainable AI (XAI), Retrieval-Augmented Generation (RAG), and AI-based applications. Successfully delivered multiple real-world projects, including the development of predictive healthcare solutions using real clinical data and production-oriented AI systems from data acquisition and model development to deployment and automation. Passionate about continuous learning, solving complex business problems through AI, and building scalable, high-impact solutions while growing as an AI/ML Engineer or Data Scientist in a fast-paced, innovation-driven environment.

TECHNICAL SKILLS

- **Programming & Data Science:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL, MS-Excel, CI/CD pipelines, ML workflow automation, FastAPI.
- **Machine Learning:** Classification, Regression, Feature Engineering, Model Evaluation**,** Hyperparameter Tuning.
- **Deep Learning:** ANN, CNN, RNN, RAG, Real-Time Video Analytics, TensorFlow, Keras, and PyTorch .
- **Concepts:** Data Preprocessing, Normalization/Scaling, Predictive Modeling, Time Series Forecasting, Statistical Analysis, Model Deployment

EXPERIENCE

Data Science Intern — 10Pearls *Oct 2025 – Dec 2025*

- Developed a fully automated, serverless AQI (Air Quality Index) prediction system forecasting air quality 3 days ahead using real-world environmental datasets.
- Built and optimized end-to-end ML/DL pipelines incorporating EDA, feature selection, missing value handling, scaling, and correlation analysis.
- Trained and benchmarked multiple ML and DL models, selected best-performing model based on MAE/RMSE evaluation metrics.
- Implemented CI/CD pipelines to automate model retraining and deployment workflows, reducing manual intervention.
- Goal: Provide early air-quality warnings to support health-focused decision-making for cities and communities.

PROJECTS

RAG Scholar Assistant — NLP / Generative AI Project *June 2026 – July*

LLMs · RAG · Vector Embeddings · BM25 · Hybrid Retrieval · NLP

- Developed a Hybrid RAG-based academic assistant capable of answering complex questions from multi-document research paper corpora.
- Combined dense semantic retrieval (vector embeddings) with sparse keyword retrieval (BM25) to significantly improve information retrieval precision and recall.
- Reduced LLM hallucinations through retrieval-enhanced generation, grounding all responses in verified source documents.
- Evaluated system across 4 key metrics: retrieval relevance, answer accuracy, response efficiency, and retrieval effectiveness.
- Improved research paper comprehension and knowledge extraction for academic and research use cases.

Kidney Disease Prediction System — Final Year Project *Mar 2025 – Jan 2026*

Random Forest · XGBoost · LightGBM · SHAP · Raspberry Pi

- Built an AI-based Chronic Kidney Disease (CKD) stage prediction system trained on 60,000+ clinical records, achieving 92% classification accuracy.
- Performed comprehensive data preprocessing including missing value imputation, outlier handling, feature engineering, and EDA to maximize data quality.
- Trained and compared multiple ML models (Random Forest, XGBoost, LightGBM, Decision Trees) using cross-

validation and hyperparameter tuning.
- Integrated SHAP (Explainable AI) for model interpretability, enabling clinicians to understand key feature contributions to predictions.
- Deployed optimized model on Raspberry Pi for real-time edge inference, enabling low-cost deployment in healthcare environments.

AI-Based Traffic Violation Detection System — Computer Vision Project *Sept 2025 – Oct 2025*

YOLO · OpenCV · Object Tracking · Video Analytics

- Developed a real-time intelligent traffic monitoring system processing live CCTV video streams for vehicle detection, classification, and tracking.
- Detected and classified multiple vehicle categories (cars, motorcycles, buses, trucks) using YOLO-based object detection.
- Implemented rule-based traffic violation detection via vehicle trajectory and movement analysis.
- Automated generation of violation alerts and timestamped evidence records for law enforcement reporting.

Social Media Content Analyzer — NLP Project *Jun 2025 – Jul 2025*

NLP · Sentiment Analysis · Text Analytics · Machine Learning

- Developed an NLP-based system to analyze and classify social media content at scale.
- Performed multi-class sentiment analysis to identify positive, negative, and neutral user opinions.
- Applied text preprocessing, TF-IDF feature extraction, and ML classification techniques for automated content categorization.
- Generated actionable insights from large volumes of unstructured social media data to support trend analysis and decision-making.

EDUCATION

Bachelor of Science in Software Engineering — Iqra University, Karachi *2022 – 2026*

- CGPA: 3.62 / 4.00

Intermediate (Pre-Engineering) — Sir Syed College, Karachi *2019 – 2021*

- Score: 82.3%

Matriculation — Hill Crest School System, Karachi *2017 – 2019*

- Score: 82.3%

CERTIFICATIONS & AWARDS

- Foundations of Data Science — Google : https://coursera.org/share/e9e64008eb101bb22062b56442de2292
- 2nd Position — Inovista Tech Competition (Kidney Disease Prediction Project)
- Presenter — Simpact, Technova, See Pakistan, AI Wrapper, SheXcelerate (AI/ML Healthcare Project)
- 10th IEEP Poster Presentation — Showcased Kidney Disease Prediction System